Elizabeth Thompson

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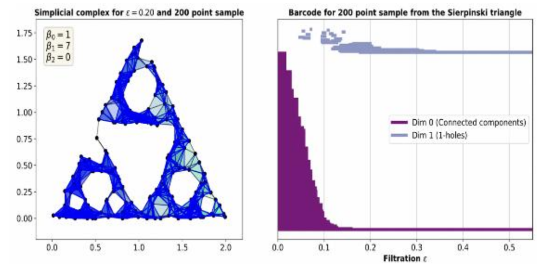
[website]

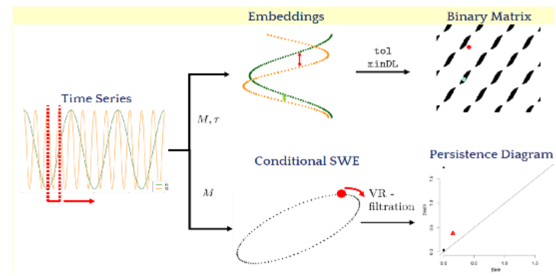
RESEARCH

Fractal Dimensions of Complex Networks: Advocating for a Topological Approach, **accepted to 3rd WinCompTop proceedings (AWM-IMA Springer volume "Research in Computational Topology")**; preprint arXiv:2506.15236



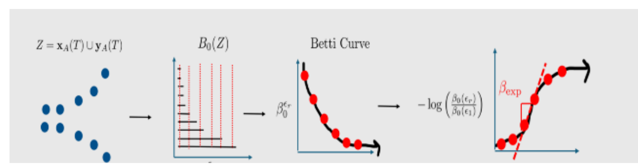
- Motivate the use of persistent homology to estimate fractal dimensions of networks through experimentally comparing the Hausdorff dimensions and dimensions obtained from persistent barcodes on increasing subsample sizes of the Sierpinski triangle (Python, R).

A Stable Measure for Conditional Periodicity of Time Series Using Persistent Homology, preprint arXiv:2501.02817



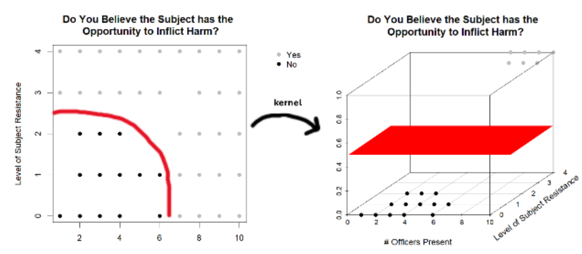
- Use persistent homology to construct a stable scoring function that quantifies how similar a given pair of time series' periodicities are (R).

A Stable Measure of Chaos in Dynamical Systems Via Persistent Homology; preprint arXiv:2601.10900



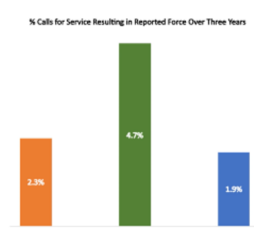
- Use persistent homology to construct a stable exponent that measures chaos in dynamical systems when compared to traditional Lyapunov methods (R).

Does Coding Change What We See? The Influence of Annotation on Perceptions of Police Use of Force; preprint CrimRxiv. [*https://doi.org/10.21428/cb6ab371.4653843d*](https://doi.org/10.21428/cb6ab371.4653843d)



- Use machine learning classification techniques to identify whether the order in which police-community interactions are observed and logically broken down impacts perceptions of severity, immediate threat, active resistance, and evasion in these encounters (Python)

Washington State Data Exchange for Public Safety (WADEPS) tutorials



- Author scripts of [short video tutorials] educating the public on finding reasonable agency comparisons and performing contextual analysis of agency patterns of behavior (R, Python, Adobe After Effects).

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TEACHING

Math 202: Calculus for Business & Economics lecture, FA 2023, WSU Vancouver

Math 103: Algebra Methods & Functions lecture, Summer 2023, WSU Vancouver

Math 108: Trigonometry lecture, Summer 2022 & 2023, WSU Vancouver

Math 140 Lab: Calculus for Life Sciences, Fall 2021 & 2023, WSU Vancouver

PRESENTATIONS

From Dynamical Systems to Criminology: Revealing Insights with Topological and Predictive Approaches, St. Anselm College, Manchester New Hampshire, April 2026 (colloquium talk)

From Dynamical Systems to Police Data: Analyzing Patterns with Topological and Machine Learning Approaches, University of Dallas, Dallas TX, February 2026 (colloquium talk)

A Stable Measure for Periodicity Similarity Using Persistent Homology, Top Time, Australian National University, Canberra AU, October 2025 (contributed talk)

What Can Donuts Tell Us About Body Cam Data? 2nd Place WSU-wide 3MT finals, WSU Pullman, March 2023 (oral)

A Stable Measure for Periodicity Similarity of Time Series Via Persistent Homology, ATMCS 11, Bozeman Montana, July 2025 (poster)

A Stable Measure of Chaos in Dynamical Systems Via Persistent Homology, Seminar in Analysis, Washington State University, September 2025 (seminar)

A Stable Measure for Periodicity Similarity of Time Series, Cascade RAIN 9, Corvallis OR, April 2025 (oral)

A Stable Measure for Conditional Periodicity of Time Series Using Persistent Homology, Joint Mathematics Meetings, January 2025 (oral)

Using Persistent Homology for Classification, WSU Vancouver, January 2025 (seminar)

Using Video Tutorials to Navigate Data Analysis in a Public Safety Open Data Exchange, American Society of Criminology, November 2024 (oral)

Perceptions of Force Severity, Operationalizing Reasonableness with a Repeated Measures Design, Western Society of Criminology, February 2024 (oral)

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AWARDS

Outstanding Graduate Student Award, Washington State University, April 2026

- The Harriet B. Rigas Award, presented to an outstanding PhD student, funded by donations from Deans and Chancellors from all WSU campuses.

WORKSHOPS & SUMMER SCHOOLS

EWM-EMS Summer School: Stability in Topological Data Analysis, Institut Mittag-Leffler, Djursholm Sweden, June 30 – July 4, 2025

- Learn about techniques for proving stability in persistent homology and work in small groups on an open problem improving stability of persistent homology classification algorithms.

Women of Computational Topology Session 3 (WinCompTop3), Lausanne Switzerland, July 2023

- Work in a small research group led by Nina Otter for one week on an applied topology project using persistent homology to estimate the fractal dimensions of complex networks.

SERVICE

Regional High School Math Contest, Washington State University, May 6 2024 and March 11, 2026

- Grade team problems & judge project presentations

AWM at SIAM Committee Member (September 1st 2027 - August 31 2029), Organizing Committee Member for AWM at SIAM Workshop (2028 Meeting)

- Recruit and communicate with organizers of the AWM Minisymposia.
- Assist in the organization of the AWM-SIAM Poster Session.
- Informed of timelines relevant to organizing SIAM activities
- Ensure that the relevant information is sent to SIAM
- Coordinate with the ED and the Managing Director

EDUCATION

PhD in Mathematics, Washington State University, May 2026

BS in Mathematics & Secondary Education, Linfield University, May 2021

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PUBLICATIONS AND PREPRINTS

Publications

Rayna Andreeva, Haydeé Contreras-Peruyero, Sanjukta Krishnagopal, Nina Otter, Maria Antonietta Pascali, and Elizabeth Thompson. Fractal dimensions of complex networks: advocating for a topological approach, 2026. arXiv:2506.15236 (to appear in the third WinCompTop proceedings AWM-IMA Springer volume "Research in Computational Topology")

Preprints

Bala Krishnamoorthy and Elizabeth Thompson. A stable measure of chaos in dynamical systems using persistent homology, 2026. arXiv:2601.10900.

Thompson, E., McMillin, M., Willits, D., Makin, D., Schulz, E., & Krishnamoorthy, B. (2025). Does Coding Change What We See? The Influence of Annotation on Perceptions of Police Use of Force. *CrimRxiv*. https://doi.org/10.21428/cb6ab371.4653843d

Bala Krishnamoorthy and Elizabeth P. Thompson. A stable measure for conditional periodicity of time series using persistent homology, 2025. arXiv:2501.02817